

- 5 (a) Equipotential lines are closer together near the cable but further apart away from cable. B1
 Since electric field strength = potential gradient($E = dV/dx$), so E decreases with distance B1
- (b) $E = dV/dx = (200-195) \times 10^3/0.0050$ M1
 $= 1.0 \times 10^6 \text{ Vm}^{-1}$ A1
- (c) Gain in ke = loss in pe M1
 $\frac{1}{2} mv^2 = eV$
 $v = \sqrt{2eV/m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 1500 / 9.11 \times 10^{-31}}$
 $= 2.30 \times 10^7 \text{ ms}^{-1}$ A1
- (d) Correct pattern/shape/arrow direction B1
 Lines fall perpendicularly to surface of shield B1
 Some lines drawn diverging outside open end of shield B1

